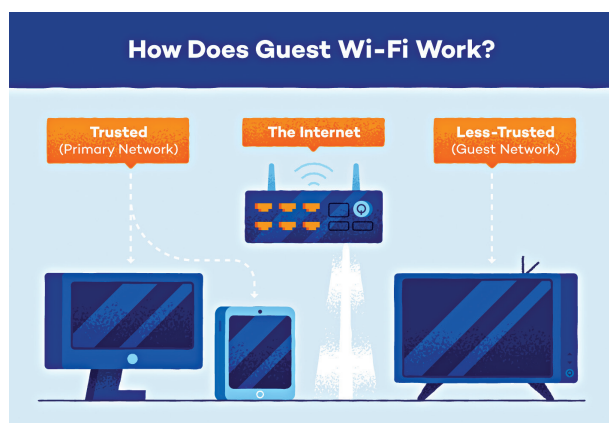




- Home_IoT
- Home_Guest

Level 2 (better): Separate IoT SSID, same LAN

If your router supports multiple SSIDs without strict guest isolation, create Home_IoT (2.4-only) but keep it on the same LAN. This often preserves local discovery while reducing band confusion.

Level 3 (best): VLAN + firewall rules (real segmentation)

If you have prosumer gear, create an IoT VLAN and apply simple rules:

- IoT → internet allowed
- IoT → main devices blocked
- main devices → IoT allowed for control (or allow only what you need)

Cybersecurity company Bitdefender's segmentation walk-through describes creating VLANs for usage types (IoT, guest, personal) and assigning devices accordingly.

Step-by-step: sanity-test your IoT lane (so you don't "break" control)

After you move devices to an IoT lane, do these checks:

1. From your phone on Home_Main, can your main smart-home app still discover and control devices?
2. Can speakers still cast/stream the way your family uses them?
3. Do automations trigger when your phone is not on the same lane? (They should – if your controller is doing its job.)
4. If something fails only on guest Wi-Fi, move that device to Level 2 (IoT SSID on the same LAN) or adjust guest isolation settings if your router allows it.

Quick sanity check: if Level 1 works and your home is happy, stop there.

7) Make devices stay connected

After coverage + lanes are set, reduce common drop-offs:

- Update router/mesh firmware regularly.
- Don't hide SSIDs during setup (some IoT setup flows fail).
- Avoid exotic password characters for picky devices.
- Prefer WPA2/WPA3 mixed mode if WPA3-only blocks older IoT.
- Keep 2.4 IoT devices on 2.4; 2.4 generally reaches farther through walls than 5 GHz.

Also: don't add nodes "just because." Every extra mesh hop can cost performance; add nodes only to eliminate real Red zones, not to chase bigger numbers on a speed test.

8) Beat power cuts with a small UPS plan

Power drops shouldn't wipe your network. When routers reboot, devices rejoin slowly and automations can miss triggers.

What should be on backup power (minimum viable smart home)

1. Modem/ONT (fiber box)
 2. Main router/mesh node
 3. Smart home hub/controller (if you have one)
- Optional (nice-to-have):
4. One extra mesh node (if your main coverage depends on it)
- Routers are typically low draw; EnergySage estimates most Wi-Fi routers use about 5–20W depending on model.

Step-by-step: size a UPS without getting lost in math

1. List devices to protect (ONT + router + hub).
2. Use the adapter label to estimate watts (e.g., $12V \times 1A = 12W$ max).
3. Add watts, then add 20–25% buffer.
4. Decide runtime:
 - 30–60 minutes for short outages and brownouts
 - 2–4 hours if your area has longer cuts
5. Use a reputable selector/runtime tool to choose a UPS.

Schneider Electric/APC provides a UPS selector tool that distinguishes watts (real power) vs VA (apparent power), helping match load to capacity. If you want the VA-to-watts relationship: Fuji Electric's guide notes $\text{watts} = \text{VA} \times \text{power factor}$.

Step-by-step: wire your network core to survive outages

1. Plug UPS into a properly earthed wall socket.
2. Plug in ONT → router → hub (in that order).
3. Simulate an outage: switch off mains and confirm Wi-Fi stays up.
4. After mains returns, confirm the internet recovers without manual reboots.

Pro tip: once a month, do a 2-minute "pull the plug" UPS test. Batteries age; the test tells you early. 